- Ass 4 due Fri 30 Oct @ 4pm

Recap: (1) A test of independence tests the null hypothesis
$H_0: p_{ij} = p_{i\cdot} \times p_{\cdot j}$ for all cells i & j

A test of homogeneity tests the null hypothesis
$H_0: f_j = f_{j'}$ for all (sub) populations j & j'

(2) Both tests are carried out in exactly the same way

Example 8.6 (continued)

OS	Faculty: Science		Engineering		Arts	Total
Windows	12	(12.51)	17	(13.9)	9 (11.58)	38
Mac	8	(10.54)	9	(11.71)	15 (9.75)	32
Linux	7	(3.95)	4	(4.39)	1 (3.66)	12
Total	27		30		25	82

Q: Is OS preference homogeneous (i.e, the same) across all 3 faculties

Null hypothesis $H_0: \underset{\sim}{p}_{Science} = \underset{\sim}{p}_{Eng} = \underset{\sim}{p}_{Arts} = \underset{\sim}{p}_{common}$

vs Alt. hypothesis H_1: no restrictions on each probability vector.

How to find expected counts under null?
Under the null, the MLE for the common vector of probabilities?

$$\hat{p}_W = \frac{38}{82}, \quad \hat{p}_M = 32/82 \quad , \hat{p}_L = 12/82$$

So for Science staff, we would expect

$27 \times 38/82 = 12.51$ to prefer Windows
$27 \times 32/82 = 10.54$ to prefer Mac
$(27 \times 12/82 = 3.95$ to prefer Linux$)$

For Engineering staff, we would expect
$30 \times 38/82 = 13.90$ to prefer Windows
$30 \times 32/82 = 11.71$ to prefer Mac.

So, Pearson's χ^2 statistic:

$$\chi^2 = \frac{(12-12.51)^2}{12.51} + \frac{(17-13.9)^2}{13.9} + \frac{(9-11.58)^2}{11.58}$$
$$+ \frac{(8-10.54)^2}{10.54} + + +$$
$$+ + + \frac{(1-3.66)^2}{3.66}$$
$$= 9.66 \text{ on } df = 4 \quad ((c-1)\times(r-1))$$

To find the p-value:



$$P(\chi^2_4 \geq 9.66) = 1 - \text{pchisq}(9.66, 4)$$
$$= 0.047 = 4.7\%.$$

Conclusion: there is moderate evidence that computer OS preference differs amongst the 3 faculties.

Any caveats for using Pearson's χ^2 here? (Note that the limiting ~~dist~~ χ^2 dist^n is a generalisation of the normal approx to a binomial proportion)

Recall that for a binomial, the normal approximation holds if we have at least 10 "successes" and at least "10 failures"

For Pearson's χ^2 test, we need to both relax & tighten this requirement:

- at least 80% cells in our table to have ≥ 10 counts
- all cells must have ≥ 5 counts.

In such cases where this fails, we may consider using Fisher's exact test

Fisher's exact test involves the following 3 logical arguments:

Step (1) Argue that the margins of the table can be considered fixed / given under the null hypothesis.
For example, the number of staff from each faculty that needed a new computer last year is determined completely by when they started at UG & has nothing to do with computer preference. So the column totals can be considered fixed
Conversely, if computer preference is NOT related to Faculty, then the 38 staff who chose Windows would have chosen Windows anyway regardless of which Faculty they are from. Similarly for the 32 Mac users & the 12 Linux users.
So, the row totals can be considered fixed under the null

Step ② Under the null, any configuration of counts within the table with the same fixed row & column totals would be equally likely to be observed.

Eg.

	Faculty									
OS	Science			Eng			Arts			Total
W	12	13	15	17	14	23	9	11	0	38
M	8	11	0	9	11	7	15	10	25	32
L	7	3	12	4	5	0	1	4	0	12
Total	27			30			25			82

Under the null, the red configuration of counts would be another, equally likely, set of possible outcomes. This configuration would have been "less unusual" than our observed counts.

Under the null, the green configuration of counts would be another, equally likely set of possible outcomes. This would have "more unusual" than our observed counts.

Step ③ By simulations / full enumerations of (all) tables of counts with a given set of row & column totals, we can determine how often we observed a set of counts "as unusual or more unusual" as our observed counts.

This is Fisher's exact test. This generally cannot be done by hand, & so we need to use a computer.

R DEMO



Final warning about categorical data: categories are often coded as numbers in a spreadsheet ... BUT those numbers are just labels & have NO numerical meaning

Eg: 1 = AUDI, 2 = BMW, 3 = BUICK, 4 = Chrysler, 5 = Citroen ...

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§10 Comparing groups using quantitative data

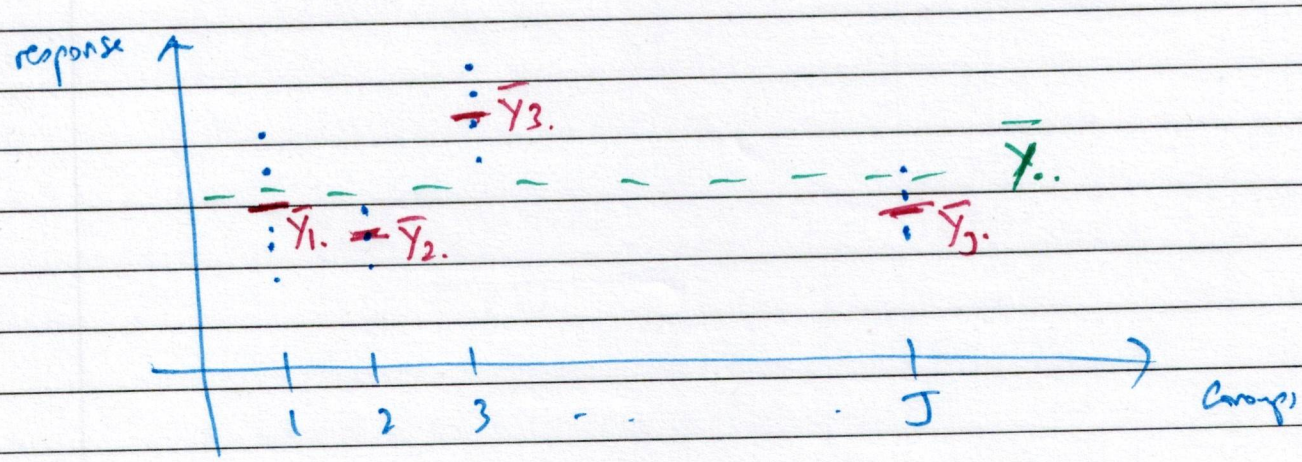
ANOVA (Analysis-of-Variance) is used to compare a quantitative variable across levels of one (or more) factor(s). ANOVA is a special case of a linear model, which we will return to later (in §9).

ANOVA setup (extension of a two-sample t-test for ≥ 2 groups)

start with σ^2 known

$$\left.\begin{array}{ll} Y_{11}, Y_{12}, Y_{13}, \ldots, Y_{1n_1} & \overset{iid}{\sim} N^*(\mu_1, \sigma^2) \\ \text{indep of } Y_{21}, Y_{22}, Y_{23}, \ldots, Y_{2n_2} & \overset{iid}{\sim} N^*(\mu_2, \sigma^2) \\ \text{indep of } Y_{31}, Y_{32}, Y_{33}, \ldots, Y_{3n_3} & \overset{iid}{\sim} N^*(\mu_3, \sigma^2) \\ \vdots & \\ \text{indep } Y_{J1}, Y_{J2}, \ldots, Y_{Jn_J} & \overset{iid}{\sim} N^*(\mu_J, \sigma^2) \end{array}\right\} \text{coming from } J \text{ (sub) pop'ns}$$

Graphically, the data may look like this:



What would be the default null hypothesis to test here?

All J groups have the same mean

$$H_0: \mu_1 = \mu_2 = \mu_3 = \cdots = \mu_J \quad (= \mu_{\text{common}})$$

vs H_1: at least one group mean is different

OR NOT all group means are the same

How do we generalize the 2-sample t-test to more than 2 groups? We can apply the GLRT principle:

$$\Lambda = \frac{\sup_{H_0} L(\mu_1, \mu_2, \ldots, \mu_J \mid \underline{Y})}{\sup_{H_1} L(\mu_1, \mu_2, \ldots, \mu_J \mid \underline{Y})}$$

In this case, this is comparing how much better the red fit is compared to the green fit for our data.

First, what is the joint likelihood for:

likelihood of first sample

$$L(\mu_1, \mu_2, \ldots, \mu_J \mid \underline{Y}) = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^{n_1} \exp\left\{-\frac{1}{2\sigma^2} \sum_{i=1}^{n_1} (Y_{1i} - \mu_1)^2\right\}$$

likelihood of second sample

$$\times \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^{n_2} \exp\left\{-\frac{1}{2\sigma^2} \sum_{i=1}^{n_2} (Y_{2i} - \mu_2)^2\right\}$$

$$\times \vdots$$

likelihood of Jth sample

$$\times \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^{n_J} \exp\left\{-\frac{1}{2\sigma^2} \sum_{i=1}^{n_J} (Y_{Ji} - \mu_J)^2\right\}$$

$$= \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^{N} \exp\left\{-\frac{1}{2\sigma^2} \sum_{j=1}^{J} \sum_{i=1}^{n_j} (Y_{ji} - \mu_j)^2\right\}$$

where $N = n_1 + n_2 + \cdots + n_J$ = total sample size

Under the null model, $H_0: \mu_1 = \mu_2 = \mu_3 = \dots = \mu_J = \mu_{common}$, what would be the MLE for this common mean?

$\hat{\mu}_1 = \hat{\mu}_2 = \hat{\mu}_3 = \dots = \hat{\mu}_J = \hat{\mu}_{common} = \bar{Y}_{..}$, the grand mean

where $\bar{Y}_{..} = \dfrac{\sum_{j=1}^{J} \sum_{i=1}^{n_j} Y_{ji}}{N}$

So $\sup_{H_0} L(\mu_1, \mu_2, \dots, \mu_J | \underline{Y}) = L(\bar{Y}_{..}, \bar{Y}_{..}, \dots, \bar{Y}_{..} | \underline{Y})$

$$= \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^N \exp\left\{-\frac{1}{2\sigma^2} \sum_{j=1}^{J} \sum_{i=1}^{n_j} (Y_{ji} - \bar{Y}_{..})^2\right\}$$

Under the alternative H_1, what would be the MLEs for each group mean?

$\hat{\mu}_1 = \bar{Y}_{1.}$, $\hat{\mu}_2 = \bar{Y}_{2.}$, $\dots$, $\hat{\mu}_J = \bar{Y}_{J.}$

So $\sup_{H_1} L(\mu_1, \mu_2, \dots, \mu_J | \underline{Y}) = L(\bar{Y}_{1.}, \bar{Y}_{2.}, \dots, \bar{Y}_{J.} | \underline{Y})$

$$= \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^N \exp\left\{-\frac{1}{2\sigma^2} \sum_{j=1}^{J} \sum_{i=1}^{n_j} (Y_{ji} - \bar{Y}_{j.})^2\right\}$$

where $\bar{Y}_{j.} = \dfrac{\sum_{i=1}^{n_j} Y_{ji}}{n_j}$, the sample mean of group j

Let's look at the sum inside the exponent of the numerator:

$$\sum_{j=1}^{J}\sum_{i=1}^{n_j}(Y_{ji}-\bar{Y}_{..})^2 = \sum_{j=1}^{J}\sum_{i=1}^{n_j}(Y_{ji}-\bar{Y}_{j.}+\bar{Y}_{j.}-\bar{Y}_{..})^2$$

(a) (b)

Total Sum of Squares $= SS_{total}$

$$= \sum_{j=1}^{J}\sum_{i=1}^{n_j}(Y_{ji}-\bar{Y}_{j.})^2 + 2\sum_{j=1}^{J}(\bar{Y}_{j.}-\bar{Y}_{..})\sum_{i=1}^{n_j}(Y_{ji}-\bar{Y}_{j.}) + \sum_{j=1}^{J}\sum_{i=1}^{n_j}(\bar{Y}_{j.}-\bar{Y}_{..})^2$$

(a') (2ba) (b')

$\sum_{i=1}^{n_j}(Y_{ji}-\bar{Y}_{j.}) = 0$

$$= \sum_{j=1}^{J}\sum_{i=1}^{n_j}(Y_{ji}-\bar{Y}_{j.})^2 + \sum_{j=1}^{J} n_j(\bar{Y}_{j.}-\bar{Y}_{..})^2$$

within group SS $= SS_{within} = SS_{resids}$

between group SS $= SS_{groups}$

So the maximal likelihood under H_0 is

$$\sup_{H_0} L = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^N \exp\left\{-\frac{1}{2\sigma^2}\left[SS_{groups} + SS_{resids}\right]\right\}$$

Denominator:

$$\sup_{H_1} L = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^N \exp\left\{-\frac{1}{2\sigma^2}\left[SS_{resids}\right]\right\}$$

So $\Lambda = \dfrac{\text{numerator}}{\text{denominator}} = \exp\left\{-\frac{1}{2\sigma^2} SS_{groups}\right\}$

Small values of $\Lambda \iff$ large values of $SS_{groups} = \sum_{j=1}^{J} n_j(\bar{Y}_{j.}-\bar{Y}_{..})^2$

But, how large is "large enough"?

Under H_0,

each $\bar{Y}_{j\cdot} \sim N(\mu_{common}, \sigma^2/n_j)$ for each group $j = 1, 2, \ldots$

$$\Rightarrow \frac{\sqrt{n_j}}{\sigma}(\bar{Y}_{j\cdot} - \mu) \overset{ind}{\sim} N(0,1)$$

$$\Rightarrow \frac{n_j}{\sigma^2}(\bar{Y}_{j\cdot} - \mu)^2 \overset{ind}{\sim} \chi^2_1 \quad \text{for each grps } j = 1, 2, \ldots, J$$

$$\Rightarrow \frac{1}{\sigma^2}\sum_{j=1}^{J} n_j(\bar{Y}_{j\cdot} - \mu)^2 \sim \chi^2_J$$

$$\text{or} \quad \sum_{j=1}^{J} n_j(\bar{Y}_{j\cdot} - \mu)^2 \sim \sigma^2 \chi^2_J$$

But we don't know μ. Our best guess is $\hat{\mu} = \bar{Y}_{\cdot\cdot}$

then $$\sum_{j=1}^{J} n_j(\bar{Y}_{j\cdot} - \bar{Y}_{\cdot\cdot})^2 \sim \sigma^2 \chi^2_{J-1}$$

$$\Leftrightarrow \frac{1}{J-1}\sum_{j=1}^{J} n_j(\bar{Y}_{j\cdot} - \bar{Y}_{\cdot\cdot})^2 \sim \sigma^2 \frac{\chi^2_{J-1}}{J-1}$$

$$\Leftrightarrow \frac{1}{J-1} SS_{groups} \overset{(1)}{\sim} \sigma^2 \frac{\chi^2_{J-1}}{J-1}$$

But in practice, we also don't know σ^2. What would be our best guess of this common σ^2?

$\sim \sigma^2\chi^2_{n_1-1}$ $\quad \sim \sigma^2\chi^2_{n_2-1}$ $\quad \sim \sigma^2\chi^2_{n_J-1}$

$$S^2_{pooled} = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2 + \cdots + (n_J-1)S_J^2}{(n_1-1) + (n_2-1) + \cdots + (n_J-1)}$$

$$= \frac{\sum_{j=1}^{J}\sum_{i=1}^{n_j}(Y_{ji} - \bar{Y}_{j\cdot})^2}{N-J} = \frac{SS_{resids}}{N-J}$$

$$\overset{②}{\sim} \sigma^2 \frac{\chi^2_{N-J}}{N-J}$$

③ Surprising (or not!), SS_{groups} & SS_{resids} are independent!

(intuition: each $\bar{Y}_{j\cdot}$ is indep of S_j^2 for normal data)

Combine ①, ②, ③ together, we get

$$\frac{\frac{SS_{group}}{J-1}}{\frac{SS_{resids}}{N-J}} \sim \frac{\frac{\sigma^2\chi^2_{J-1}}{J-1}}{\frac{\sigma^2\chi^2_{N-J}}{N-J}}$$

($\hat{\sigma}^2 \nearrow \frac{SS_{resids}}{N-J}$; ① over $\frac{\sigma^2\chi^2_{J-1}}{J-1}$; ② over $\frac{\sigma^2\chi^2_{N-J}}{N-J}$; ↳ indep ③)

$$= F_{(J-1),(N-J)} \quad \text{(by def of an F dist)}$$

numerator df is the complexity (# parameters) in the alternative model minus the complexity (# parameters) in the null model

denominator df measures how accurately we can estimate σ^2 (noise)